46,XY Sex Reversal 5 (SRXY5): A Comprehensive Disease Characterization

Disease: 46,XY Sex Reversal 5 (SRXY5) **OMIM:** #613080 · **MONDO:** MONDO:0013120 · **Causal gene:** *CBX2* (17q25.3) **Category:** Mendelian, autosomal-recessive disorder/difference of sex development (DSD) **Report date:** 2026-09-01

Summary

46,XY Sex Reversal 5 (SRXY5) is a rare autosomal-recessive difference of sex development (DSD) caused by biallelic loss-of-function mutations in *CBX2* (Chromobox 2; historically *M33*), a Polycomb-group (PRC1) chromatin-modifying gene on chromosome 17q25.3. In an individual with a normal 46,XY karyotype, loss of CBX2 function prevents activation of the testis-determining genetic cascade. Because CBX2 acts *upstream of SRY*, its loss produces failure to initiate the SRY→SOX9 male pathway (with reduced NR5A1/SF-1 expression) and de-repression of the ovarian program. The clinical consequence ranges from a completely female phenotype with a uterus and histologically normal ovaries at one extreme to ambiguous/undervirilized external genitalia at the other.

The disease is defined by the index case reported by Biason-Lauber and colleagues in 2009 (PMID: 19361780): a prenatally karyotyped 46,XY girl with normal female external genitalia, a uterus, and ovaries, who carried compound-heterozygous loss-of-function mutations in *CBX2* — NM_005189.3:c.293C>T (p.Pro98Leu) and c.1328G>C (p.Arg443Pro). Both alleles are independently classified **Pathogenic** for "46,XY sex reversal 5" in ClinVar. The human finding is mechanistically anchored by decades of mouse work: constitutive *M33/Cbx2* knockout recapitulates XY male-to-female sex reversal, and this reversal is genetically rescued by forced expression of *Sry* or *Sox9*, cementing CBX2's position at the top of the testis-determination hierarchy.

SRXY5 is exceptionally rare (only a small number of confirmed *CBX2*-mutant cases worldwide; a 47-patient DSD cohort found no additional pathogenic *CBX2* mutations). There is no disease-modifying therapy; management is supportive and multidisciplinary — sex-of-rearing decisions, hormone replacement, gonadectomy where germ-cell-tumor risk warrants, and

psychosocial and genetic counseling — following the international DSD consensus framework. This report synthesizes 9 confirmed findings drawn from 21 reviewed papers spanning human clinical, in-vitro/functional, model-organism, and computational-database evidence.

Key Findings

Finding 1 — SRXY5 is caused by biallelic loss-of-function mutations in *CBX2*, placing CBX2 upstream of SRY

The defining discovery of SRXY5 came from a single, unusually informative index case. Biason-Lauber et al. (2009) described a prenatally karyotyped 46,XY girl born with **completely normal female external genitalia, a uterus, and histologically normal ovaries**. Whole-gene analysis of *CBX2* — the human ortholog of mouse *M33* — revealed compound-heterozygous loss-of-function mutations. This case defines the OMIM entry #613080 (46,XY sex reversal 5).

Crucially, the discovery added a new, higher tier to the human sex-determination cascade. As the authors state:

> "The analysis of the human homolog of M33, Chromobox homolog 2 (CBX2), in this girl revealed loss-of-function mutations that allowed us, by placing CBX2 upstream of SRY, to add an additional component to the still incomplete cascade of human sex development."
> — PMID: 19361780

This is the single most important mechanistic anchor for the disease: CBX2 is not merely another testis gene but an upstream *enabler* of the SRY-initiated switch.

Finding 2 — Mouse *M33/Cbx2* knockout recapitulates XY sex reversal via loss of upstream regulation of *Sry* and *Nr5a1*/SF-1

The human finding rests on a robust animal foundation predating it by more than a decade. Katoh-Fukui et al. (1998) showed that *M33*-null mice display **XY male-to-female sex reversal** with retarded genital ridge formation and gonadal growth defects arising near the time of *Sry* expression:

> "survivors showed male-to-female sex reversal" — PMID: 9641679
>
> "Gonadal growth defects appeared near the time of expression of the Y-chromosome-specific Sry gene, suggesting that M33 deficiency may cause sex reversal by interfering with steps upstream of Sry." — PMID: 9641679

A subsequent study connected M33 to the nuclear receptor SF-1: M33-knockout adrenal/splenic phenotypes mirror those of *Nr5a1* (Ad4BP/SF-1) knockouts, and M33-KO gonads/adrenals showed significantly reduced Ad4BP/SF-1 expression with ChIP evidence of direct regulation:

> "indicating that M33 is an essential upstream regulator of Ad4BP/SF1" — PMID: 15899914

Sex-reversal penetrance is incomplete in mice: one study reported reversal in ~28.6% of XY−/− embryos, with the remainder showing bilateral testicular hypoplasia (PMID: 22200029).

Finding 3 — CBX2 is a PRC1 chromatin regulator that *stimulates* the male pathway and *represses* the female pathway; two isoforms have distinct DSD roles

Genome-wide DamID mapping in Sertoli-like NT-2D1 cells identified ~1,600 direct CBX2 targets and established CBX2's bistable, dual role:

> "CBX2 role in the sex development cascade is to stimulate the male pathway and concurrently inhibit the female pathway" — PMID: 25569159

Mechanistically, CBX2 is an H3K27me3 "reader" within Polycomb Repressive Complex 1 (PRC1), where it associates with RING1B, PCGF2, and PHC2 and plays a structural role in the H2AK119 mono-ubiquitination machinery (PMID: 31093962; PMID: 32979540).

The gene encodes two functionally distinct isoforms. The shorter isoform, **CBX2.2**, has its own DSD-associated variants (p.Cys132Arg and p.Cys154fs) that cause 46,XY DSD, likely through defective regulation of *EMX2*:

> "both CBX2.2 variants fail to regulate the expression of genes essential for sexual development, leading to a severe 46,XY DSD defect, likely because of a defective expression of EMX2 in the developing gonad" — PMID: 29998616

Finding 4 — Phenotypic spectrum ranges from complete female genitalia with ovaries to ambiguous genitalia

SRXY5 is phenotypically variable. The index case had a fully female phenotype with uterus and normal ovaries (PMID: 19361780). At the opposite end, CBX2.2-variant patients presented with:

> "two patients with features of DSD i.e. atypical external genitalia, perineal hypospadias and no palpable gonads" — PMID: 29998616

Because 46,XY gonadal dysgenesis broadly carries elevated germ-cell-tumor/gonadoblastoma risk, gonadectomy is considered, and care follows the multidisciplinary DSD consensus framework:

> "medical, surgical and psychological care and the decision regarding sex of rearing or gender assignment" — PMID: 18987491

Finding 5 — Gene annotation and gnomAD constraint support an autosomal-recessive mechanism

CBX2 identifiers: **HGNC:1552**, NCBI Gene **84733**, Ensembl **ENSG00000173894**, UniProt **Q14781**, OMIM gene ***602770**, cytoband** 17q25.3 **(chr17:79,778,135–79,788,394, GRCh38); aliases *M33*, *CDCA6*, *SRXY5*. gnomAD constraint metrics show CBX2 is** not **loss-of-function intolerant: pLI = 0.024, observed/expected LoF (oe_lof) = 0.52 (90% CI 0.33–0.87), lof_z = 1.85. Heterozygous LoF is therefore tolerated in the general population, consistent with a** recessive *disease requiring biallelic loss. ClinVar lists 224 CBX2* variants, predominantly benign/VUS.

Finding 6 — CBX2 acts upstream of SRY: rescue epistasis with *Sry/Sox9*

The decisive genetic proof of hierarchy comes from mouse rescue experiments. In *Cbx2*-KO gonads, expression of *Sry, Sox9, Lhx9, Ad4BP/SF-1 (Nr5a1), Dax-1 (Nr0b1), Gata4, Arx,* and *Dmrt1* is disrupted:

> "the expression of Sry, Sox9, Lhx9, Ad4BP/SF-1, Dax-1, Gata4, Arx, and Dmrt1, genes encoding transcription factors essential for gonadal development, is affected in Cbx2 KO gonads" — PMID: 22186409

Forced expression of *Sry* or *Sox9* rescues the sex reversal but not the gonadal hypoplasia:

> "Male-to-female sex reversal in Cbx2 KO mice was rescued by crossing them with transgenic mice displaying forced expression of Sry or Sox9." — PMID: 22186409

This dissociates two CBX2 functions: (i) controlling the sex-determining switch *specifically through Sry*, and (ii) governing gonad size via a separate downstream gene set.

Finding 7 — Disease and molecular ontology annotations

Disease IDs: OMIM #613080; MONDO:0013120 (confirmed via EBI OLS); Orphanet groups under 46,XY complete/partial gonadal dysgenesis (ORPHA:242, ORPHA:2138). **Gene/protein:** *CBX2* (HGNC:1552, OMIM *602770, UniProt Q14781, 532 aa). Protein architecture (UniProt Q14781): N-terminal chromodomain (aa 12–70; H3K9me/H3K27me reader), nuclear localization signal (aa 163–168), and large disordered/AT-hook-containing C-terminal regions (aa 60–204, 296–348, 379–493). Notably, the index-case variants p.Pro98Leu and p.Arg443Pro fall in the disordered regions* outside* the chromodomain.

Suggested GO terms: GO:0035102 (PRC1 complex), GO:0031519 (PcG protein complex), GO:0045137 (development of primary sexual characteristics), GO:0062072 (histone reader activity), GO:0000122/GO:0045892 (negative regulation of transcription), GO:0031507 (heterochromatin formation), GO:0005634 (nucleus).

Finding 8 — ClinVar confirms the two index-case alleles as Pathogenic

ClinVar (queried 2026-09-01) lists **NM_005189.3(CBX2):c.293C>T (p.Pro98Leu)** and **NM_005189.3(CBX2):c.1328G>C (p.Arg443Pro)**, both classified **Pathogenic** for the condition "46,XY sex reversal 5" — precisely the compound-heterozygous genotype of the index patient (PMID: 19361780). Other *CBX2* coding substitutions (p.His331Arg, p.Met404Leu, p.Val487Ile, p.Ala460Thr) are VUS. Pathogenic large 17q25.3 CNVs in ClinVar are contiguous-gene events rather than isolated CBX2-DSD.

Finding 9 — Mouse ortholog identifiers and completed model-organism annotation

Mouse ortholog *Cbx2* (*M33*): NCBI Gene **12416**, MGI:**88289**, Ensembl **ENSMUSG00000025577**, chromosome 11. The constitutive *M33/Cbx2* knockout is the validated SRXY5 model recapitulating XY male-to-female sex reversal (PMID: 9641679, PMID: 22186409, PMID: 15899914, PMID: 22200029).

Section-by-Section Report

1. Disease Information

Overview. SRXY5 is a rare Mendelian 46,XY DSD in which a chromosomally male (46,XY) individual fails to complete testis determination because of loss of the Polycomb regulator CBX2. The result is dysgenetic gonads or ovaries, frequently with Müllerian (uterine) structures, and external genitalia ranging from typically female to ambiguous.

Key identifiers:

Resource	Identifier
OMIM (phenotype)	#613080
OMIM (gene)	*602770 (CBX2)
MONDO	MONDO:0013120
Orphanet (grouping)	ORPHA:242, ORPHA:2138 (46,XY complete/partial gonadal dysgenesis)
Gene (HGNC)	HGNC:1552
NCBI Gene	84733
Ensembl	ENSG00000173894
UniProt	Q14781
MeSH (broad)	Disorders of Sex Development; Gonadal Dysgenesis, 46,XY

Synonyms / alternative names: 46,XY sex reversal 5; SRXY5; CBX2-related 46,XY DSD; gonadal dysgenesis due to CBX2 deficiency. Gene aliases: *M33*, *CDCA6*.

Information source. Disease-level knowledge derives primarily from aggregated resources (OMIM, ClinVar, Orphanet) built on a very small number of *individual* clinical case reports plus extensive mouse model data — not from EHR-scale datasets.

2. Etiology

Primary cause — genetic. SRXY5 is caused by **biallelic (autosomal-recessive) loss-of-function mutations in *CBX2***. The index genotype is compound-heterozygous p.Pro98Leu / p.Arg443Pro (PMID: 19361780). Isoform-specific CBX2.2 variants (p.Cys132Arg, p.Cys154fs) cause a severe 46,XY DSD via defective *EMX2* regulation (PMID: 29998616).

Genetic risk factors. The disease requires two damaging *CBX2* alleles; heterozygous carriers are unaffected (consistent with gnomAD tolerance of monoallelic LoF, pLI = 0.024). Consanguinity increases risk of biallelic recessive genotypes (a recurring theme in DSD cohorts generally, e.g., PMID: 42202777).

Environmental / infectious factors. None known to cause SRXY5. This is a monogenic developmental disorder; there is no evidence for toxin, infectious, or lifestyle etiology.

Protective factors and gene–environment interactions. Not applicable / none established. The only "protective" scenario is the absence of a second pathogenic allele.

3. Phenotypes

Phenotype	Type	Suggested HPO	Frequency / notes
46,XY complete gonadal dysgenesis / sex reversal	Clinical sign	HP:0010461 (46,XY sex reversal)	Core; variable
Female external genitalia in 46,XY individual	Physical manifestation	HP:0000812 (abnormal external genitalia)	Index case
Ambiguous genitalia	Physical manifestation	HP:0000062	CBX2.2-variant patients
Hypospadias (perineal)	Physical manifestation	HP:0000047	CBX2.2 cases (PMID: 29998616)
Presence of uterus / Müllerian derivatives	Clinical sign	HP:0000130 (abnormal uterus)	Index case had uterus
Ovarian or dysgenetic gonadal tissue	Histology	HP:0000138 / HP:0000133	Index case: normal ovaries
Cryptorchidism / no palpable gonads	Clinical sign	HP:0000028	CBX2.2 cases
Germ-cell tumor / gonadoblastoma risk	Neoplasm (risk)	HP:0100728 / HP:0100729	Elevated in 46,XY GD generally
Primary amenorrhea / delayed puberty (potential)	Lab/clinical	HP:0000132 / HP:0000823	Depends on gonadal function

Characteristics. Onset is **congenital** (determined during embryonic gonadal development, ~gestational weeks 6–8 in humans). Severity is **variable** (from typically female to ambiguous). Course is **stable/non-progressive** structurally, though tumor risk accrues over time and pubertal hormone deficits emerge with age. Frequency data are limited by the very small number of confirmed cases.

Quality-of-life impact. DSD conditions carry documented psychosocial and quality-of-life burdens; a multidisciplinary education/empowerment program (Empower-DSD) improved or stabilized health-related quality of life in >66% of children and parents (PMID: 42597469) and improved diagnosis-specific knowledge (PMID: 41579703).

4. Genetic / Molecular Information

Causal gene. *CBX2* (HGNC:1552; OMIM *602770; 17q25.3).

Pathogenic variants (ClinVar, Pathogenic for SRXY5):

Variant (NM_005189.3)	Protein	Type	Classification	Domain location
c.293C>T	p.Pro98Leu	Missense	Pathogenic	Disordered region (outside chromodomain)
c.1328G>C	p.Arg443Pro	Missense	Pathogenic	Disordered C-terminal region
(CBX2.2)	p.Cys132Arg	Missense	Reported pathogenic (PMID: 29998616)	Isoform-specific
(CBX2.2)	p.Cys154fs	Frameshift	Reported pathogenic	Isoform-specific

Allele frequency. Pathogenic alleles are ultra-rare/private; gnomAD shows CBX2 is LoF-tolerant at the heterozygous level (oe_lof 0.52). **Origin:** germline. **Functional consequence:** loss of function (biallelic).

Modifier genes / epigenetics. CBX2 itself is an epigenetic effector (H3K27me3 reader, H2AK119ub machinery). Downstream network members (*SRY, SOX9*, *NR5A1/SF-1*, *EMX2*, *DMRT1*, *GATA4*, *DAX1/NR0B1*, *LHX9*, *ARX*) are candidate modifiers of expressivity. No formal modifier locus is established.

Chromosomal abnormalities. Large 17q25.3 CNVs in ClinVar are contiguous-gene deletions/duplications, not isolated CBX2-DSD events. A 47-patient DSD MLPA study found **no** CBX2 copy-number changes and no additional pathogenic point mutations (PMID: 23219007), underscoring rarity.

5. Environmental Information

Not applicable. SRXY5 is a monogenic developmental disorder with no established environmental, lifestyle, toxicological, or infectious contribution.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic loss-of-function mutation in *CBX2*** (e.g., p.Pro98Leu + p.Arg443Pro) **results in** a non-functional CBX2 protein in the bipotential gonad (genital ridge). *(Demonstrated — human PMID: 19361780; ClinVar Pathogenic.)*
2. Loss of CBX2 **impairs assembly/function of the PRC1 chromatin-modifying complex** (CBX2 is a structural H3K27me3 reader supporting H2AK119 mono-ubiquitination). *(Demonstrated in vitro — PMID: 31093962, PMID: 32979540.)*
3. Defective CBX2/PRC1 activity **fails to establish the chromatin state that stimulates male-pathway genes and represses female-pathway genes** across ~1,600 direct targets. *(Demonstrated — DamID, PMID: 25569159.)*
4. As an upstream node, CBX2 loss **fails to permit/activate *SRY* (and *NR5A1*/SF-1) expression** in pre-Sertoli somatic cells at the critical window (~gestational week 6–7; in mouse, near the time of *Sry* onset). *(Demonstrated — mouse PMID: 9641679, PMID: 15899914, PMID: 22186409.)*
5. Absent SRY **fails to activate *SOX9***, the master Sertoli-cell determinant. **Branch:** without SOX9-driven Sertoli differentiation, the supporting-cell lineage defaults toward the granulosa (ovarian) program; de-repressed pro-ovarian genes (e.g., via impaired *EMX2* regulation in the CBX2.2 isoform axis) reinforce this. *(Demonstrated/inferred — PMID: 22186409, PMID: 29998616.)*
6. Failure of Sertoli-cell determination **results in gonadal dysgenesis or ovarian development** in a 46,XY gonad. *(Demonstrated — human/mouse.)*
7. Absent/deficient testicular Sertoli and Leydig function **leads to loss of anti-Müllerian hormone and androgen output**, which **results in** retention of Müllerian structures (uterus) and undervirilized/female external genitalia. *(Inferred from endocrine physiology; consistent with index-case uterus + female genitalia.)*
8. In parallel, a **separate CBX2-dependent gene set governs gonad size**, so dysgenetic/hypoplastic gonads persist even when the sex-fate switch is rescued. *(Demonstrated — Sry/Sox9 rescue corrects sex reversal but not hypoplasia, PMID: 22186409.)*

```
CBX2 biallelic LOF
        │ (results in)
        ▼
PRC1/H2AK119ub chromatin regulation impaired
        │ (fails to set)
        ▼
Male genes not stimulated / female genes de-repressed
        │ (fails to activate)
        ▼
SRY not expressed ──► SOX9 not activated
        │                         │
        │ (branch)                ▼
        ▼                 Sertoli differentiation fails
Pro-ovarian program             │
(EMX2 axis) reinforced          ▼
        └─────────► Gonadal dysgenesis / ovary in 46,XY
                                 │ (leads to)
                                 ▼
        ↓ AMH, ↓ androgens ► uterus retained + female/ambiguous genitalia
                                 │
        (parallel) separate gene set ► persistent gonadal hypoplasia
```

Upstream vs downstream. CBX2 is the most **upstream** demonstrated node (above SRY). SRY→SOX9→Sertoli differentiation and the NR5A1/SF-1, DMRT1, GATA4, DAX1, LHX9, ARX network are **downstream**.

Cell types / processes. Key cell type: bipotential/pre-Sertoli somatic supporting cell of the genital ridge (**suggested CL:** CL:0000216 Sertoli cell; CL:0000670 primordial germ cell; CL:0000501 granulosa cell). Processes: **GO:0007530** sex determination, **GO:0008584** male gonad development, **GO:0045137** development of primary sexual characteristics, **GO:0031507** heterochromatin formation.

Molecular profiling. DamID identified ~1,600 direct CBX2 targets in Sertoli-like cells (PMID: 25569159). No SRXY5-specific human metabolomic/proteomic/single-cell datasets are available given case rarity.

7. Anatomical Structures Affected

- **Primary organ:** gonad (bipotential gonad / genital ridge) — **UBERON:0000991** gonad; **UBERON:0000992** ovary; **UBERON:0000473** testis.
- **Secondary / body systems:** reproductive/endocrine system (**UBERON:0000990**); Müllerian derivatives — uterus **UBERON:0000995**, external genitalia. Adrenal gland and spleen are

affected in the mouse M33 model (via SF-1) but adrenal disease is not a prominent human SRXY5 feature.

- **Tissue/cell level:** gonadal somatic supporting cells (Sertoli/granulosa) and germ cells; germline meiotic defects in the mouse model (PMID: 22200029).
- **Subcellular:** nucleus / chromatin (**GO:0005634** nucleus; **GO:0035102** PRC1 complex; **GO:0000785** chromatin).
- **Lateralization:** typically bilateral gonadal involvement; asymmetric gonads (one hypoplastic testis + contralateral ovary/ovotestis) occur in the mouse model.

8. Temporal Development

- **Onset:** congenital — the lesion acts during embryonic sex determination (human ~6–8 weeks gestation).
- **Onset pattern:** insidious/developmental; often first detected at birth (ambiguous genitalia), prenatally (karyotype–phenotype discordance, as in the index case), or at puberty (delayed puberty/primary amenorrhea).
- **Progression:** structurally stable/non-progressive; lifelong. Germ-cell-tumor risk accrues over time, and hypogonadism manifests at expected puberty.
- **Critical period:** the sex-determination window is the intervention-relevant developmental window mechanistically, though no in-utero therapy exists.

9. Inheritance and Population

- **Inheritance:** autosomal recessive (biallelic *CBX2* LoF). Consistent with gnomAD LoF tolerance (pLI 0.024).
- **Prevalence/incidence:** not precisely established — ultra-rare; only a handful of molecularly confirmed cases worldwide. A dedicated 47-patient DSD cohort found no additional pathogenic CBX2 mutations (PMID: 23219007), and the authors concluded the study "does not support CBX2 gene disruption as a common cause of gonadal DSD."
- **Penetrance/expressivity:** variable expressivity (female to ambiguous phenotype); mouse penetrance of overt sex reversal is incomplete (~28.6% of XY−/−, PMID: 22200029).
- **Consanguinity/founder effects:** consanguinity raises recessive-genotype probability (general DSD context); no CBX2 founder allele is established.
- **Carrier frequency:** unknown; individual pathogenic alleles are private/ultra-rare in gnomAD.

- **Sex ratio / affected population:** by definition affects 46,XY (chromosomally male) individuals who may present or be reared as female; no ethnic predilection established.

10. Diagnostics

Recommended approach: karyotype (46,XY) with phenotype–karyotype discordance triggers molecular workup.

- **Genetic testing:** karyotyping (confirms 46,XY); DSD **gene panels** including *CBX2* alongside *SRY, SOX9, NR5A1, MAP3K1, WT1, GATA4, DHH, DMRT1*; **WES/WGS** for gene-agnostic diagnosis (essential given rarity); **single-gene *CBX2* sequencing** for targeted confirmation; **chromosomal microarray/MLPA** to detect CBX2 CNVs (MLPA probe set developed in PMID: 23219007). Confirm variants against **ClinVar** (both index alleles are Pathogenic).
- **Laboratory tests:** gonadotropins (LH/FSH), testosterone, AMH, inhibin B, hCG stimulation test to assess gonadal function.
- **Imaging:** pelvic/abdominal ultrasound and MRI to identify Müllerian structures (uterus) and locate gonads.
- **Histopathology:** gonadal biopsy — ovarian, dysgenetic, or streak tissue; surveillance for gonadoblastoma/germ-cell neoplasia.
- **Differential diagnosis:** other 46,XY complete/partial gonadal dysgenesis (SRY, NR5A1/SF-1, MAP3K1, WT1, SOX9, DHH, PBX1), androgen insensitivity syndrome, 17α-/11β-hydroxylase and steroidogenic defects, and Swyer syndrome. Distinguishing features: *CBX2*-related cases show early upstream failure with possible normal ovaries; PBX1 disease adds radioulnar/radiocubital synostosis (PMID: 31058389).
- **Screening:** no population newborn screen; cascade genetic testing of relatives after a proband is identified.

11. Outcome / Prognosis

- **Survival/mortality:** SRXY5 is not intrinsically life-limiting; life expectancy is normal. Principal medical risk is **germ-cell tumor/gonadoblastoma** in dysgenetic gonads, mitigated by surveillance/gonadectomy.
- **Morbidity/function:** infertility is typical; hypogonadism requires lifelong hormone replacement; psychosocial burden is significant.
- **Complications:** gonadal neoplasia; osteoporosis and metabolic effects of untreated hypogonadism; surgical and psychological sequelae.

- **Prognostic factors:** phenotype severity, gonadal histology/position, and timing of diagnosis/intervention. No molecular prognostic biomarker beyond genotype is established.
- **Quality of life:** improvable with structured multidisciplinary support (PMID: 42597469, PMID: 41579703).

12. Treatment

No disease-modifying/curative therapy exists (no gene, cell, or RNA therapy; not applicable). Management is supportive and individualized within the international DSD consensus framework (PMID: 16882788, PMID: 18987491, PMID: 17885459):

Modality	Detail	Suggested NCIT
Hormone replacement	Estrogen (± progestin) or testosterone per sex of rearing/gonadal status	NCIT:C15417 (hormone therapy)
Gonadectomy	For germ-cell-tumor risk in dysgenetic gonads	NCIT:C15277 (surgery) / gonadectomy
Genital / reconstructive surgery	Individualized; deferred until autonomous consent where legally required (e.g., Germany)	NCIT:C15329 (reconstructive surgery)
Psychological support	DSD-specialized counseling; peer/empowerment programs	supportive care
Genetic counseling	Recurrence-risk (25% for AR), cascade testing	NCIT:C15681 (genetic counseling)

Sex-of-rearing decisions should be based on the most likely adult gender identity, diagnosis, genital appearance, fertility potential, and psychosocial context (PMID: 17885459). No pharmacogenomic or experimental targeted therapy is specific to SRXY5.

13. Prevention

- **Primary prevention:** not possible for a congenital monogenic disorder; **genetic counseling** and reproductive options (PGT/prenatal testing) for at-risk (carrier) couples.
- **Secondary prevention:** early molecular diagnosis; gonadal tumor **surveillance** and timely gonadectomy where indicated.

- **Tertiary prevention:** hormone replacement to prevent osteoporosis/metabolic complications; psychosocial support to reduce stigma-related morbidity.
- **Counseling:** autosomal-recessive recurrence risk 25% per pregnancy for carrier couples; cascade carrier testing. Immunization/public-health/environmental measures: not applicable.

14. Other Species / Natural Disease

- **Taxonomy / ortholog:** mouse *Cbx2* (NCBI Gene 12416, MGI:88289, Ensembl ENSMUSG00000025577, chromosome 11); NCBI Taxon 10090 (*Mus musculus*).
- **Natural disease:** no well-documented naturally occurring CBX2 sex-reversal disease in companion animals/wildlife is recorded here; the phenotype is known from engineered knockouts. (OMIA not confirmed for a spontaneous CBX2 DSD in this investigation.)
- **Comparative biology:** the CBX2→Sry/SF-1 hierarchy is conserved between mouse and human, validating cross-species inference; mouse recapitulates XY male-to-female sex reversal and gonadal hypoplasia.
- **Zoonotic potential:** not applicable.

15. Model Organisms

- **Primary model:** constitutive *M33/Cbx2*-knockout mouse (mammalian; Alliance/MGI:88289).
- **Model types available:** knockout (constitutive); the field also uses transgenic rescue lines (forced *Sry* or *Sox9* expression) demonstrating epistasis (PMID: 22186409). In-vitro human models: NT-2D1 Sertoli-like cells (DamID target mapping, PMID: 25569159); gonadal fibroblasts and EBV-transformed lymphocytes express both CBX2 isoforms (PMID: 23219007).
- **Phenotype recapitulation:** high — XY male-to-female sex reversal, gonadal hypoplasia, reduced SF-1, disrupted downstream TF network; additional homeotic skeletal transformations and germline/meiotic defects (broader Polycomb roles).
- **Limitations:** incomplete penetrance (~28.6%); pleiotropic phenotypes (skeletal, splenic, adrenal) beyond the human DSD focus; isoform biology (CBX2.1 vs CBX2.2) is human-relevant and less fully modeled in mouse.
- **Resources:** MGI, IMPC/IMSR for allele availability.

Mechanistic Model / Interpretation

SRXY5 is best understood as a **failure of the upstream "permissive" chromatin switch** that normally licenses the male genetic program. CBX2, as a PRC1 reader of H3K27me3, sets the chromatin landscape that (a) *permits/activates SRY and NR5A1/SF-1* and (b) *represses the ovarian program*. Because CBX2 sits above SRY, its biallelic loss is functionally equivalent — at the level of outcome — to SRY loss, but it acts one tier higher and simultaneously de-represses female genes. The mouse rescue experiments provide the cleanest logic: restoring *Sry* or *Sox9* downstream corrects the sex-fate decision, proving CBX2's role in that decision is transmitted *through* SRY/SOX9; yet gonad size remains hypoplastic, revealing a second, parallel CBX2 output. This two-arm model (fate switch vs. growth) explains the clinical spectrum: patients can have ovaries with a uterus (fate fully flipped) or dysgenetic/ambiguous gonads (partial), depending on residual function and isoform involvement (CBX2.2→EMX2).

Node	Role	Direction	Evidence
CBX2 (PRC1)	Chromatin permissive switch	Most upstream	Human [19361780]; mouse [9641679]
SRY	Testis-determining trigger	Downstream of CBX2	Rescue [22186409]
SOX9	Master Sertoli determinant	Downstream of SRY	Rescue [22186409]
NR5A1/SF-1	Steroidogenic/gonadal TF	Downstream target	[15899914]
EMX2 (via CBX2.2)	Ovarian/gonadal regulator	Branch	[29998616]

Evidence Base

PMID	Title (abbrev.)	Evidence type	Role
19361780	Ovaries/female phenotype in 46,XY girl with CBX2 mutations	Human clinical (index case)	Defines SRXY5; CBX2 upstream of SRY
22186409	*Cbx2* required for *Sry* expression	Mouse genetic	Epistasis: Sry/Sox9 rescue
9641679	Male-to-female sex reversal in M33 mutants	Mouse	Founding model; upstream of Sry
15899914	M33 regulates Ad4BP/SF1	Mouse/ChIP	SF-1 link
25569159	Genome-wide CBX2 targets	In vitro (DamID)	Dual stimulate/ repress role
29998616	CBX2 isoform 2 targets in DSD	Human/ functional	CBX2.2 variants; EMX2
31093962	PRC1 topology/enzymology	In vitro biochemistry	CBX2 structural role in PRC1
32979540	CBX protein functions review	Review	CBX2 as H3K27me3 reader in PRC1
22200029	Cbx2 in meiosis/germline	Mouse	Penetrance ~28.6%; germline role
23219007	CBX2 in 46,XY/46,XX DSD cohort	Human cohort (n=47)	CBX2 not a common DSD cause; rarity
31058389	PBX1 in testis-determination	Human	CBX2 protein-interaction partner; DDx
18987491 / 16882788 / 17885459	DSD consensus statements	Clinical guideline	Management framework
42597469 / 41579703	Empower-DSD program	Clinical (QoL/ education)	Quality-of-life/ support evidence

Challenging/tempering evidence: PMID: 23219007 explicitly found no pathogenic *CBX2* mutations in 47 DSD patients — a key check on over-attribution: CBX2 is a *rare* cause, not a common one.

Limitations and Knowledge Gaps

1. **Extreme rarity / small N.** The human disease is defined by very few molecularly confirmed cases; genotype–phenotype correlations, penetrance, and prevalence are consequently imprecise.
2. **Mechanistic gaps in humans.** The full downstream target set and the precise chromatin logic (which male genes are directly activated vs. which female genes de-repressed) are best characterized in cell lines and mouse, not patient gonads.
3. **Isoform biology.** The distinct contributions of CBX2.1 vs CBX2.2 (and the EMX2 axis) are incompletely resolved and may explain phenotypic variability.
4. **No prevalence/epidemiology data**, no natural-history cohort, and no SRXY5-specific omics datasets.
5. **No SRXY5-specific therapy or biomarker**; management is generic to 46,XY gonadal dysgenesis.
6. **Domain location paradox.** The pathogenic index missense variants (p.Pro98Leu, p.Arg443Pro) lie outside the chromodomain in disordered regions — the structural basis of their loss of function is not fully explained.

Proposed Follow-up Experiments / Actions

1. **Establish an SRXY5 patient registry** and pursue GeneMatcher/international case aggregation to define penetrance, expressivity, and gonadal-tumor risk quantitatively.
2. **Functional characterization of p.Pro98Leu and p.Arg443Pro** (and CBX2.2 variants) in isogenic gonadal-somatic iPSC-derived models to map how disordered-region substitutions disrupt PRC1 assembly and target regulation.
3. **Single-cell / spatial transcriptomics of patient or knockout gonadal ridge** to resolve the Sertoli-vs-granulosa fate branch and validate the CBX2→SRY→SOX9 and CBX2.2→EMX2 arms in situ.

4. **Systematic reclassification of the many CBX2 VUS** in ClinVar using calibrated functional assays to improve diagnostic yield.
5. **Comparative-genomics / OMIA search** for spontaneous CBX2-related DSD in domestic species to add natural-disease models.
6. **Formalize a DSD-panel diagnostic algorithm** ensuring CBX2 (with CNV/MLPA coverage) is included and cross-referenced to ClinVar, and integrate multidisciplinary psychosocial support (Empower-DSD-type programs) into standard care.

Report compiled from 9 confirmed findings and 21 reviewed papers, integrating human clinical, in-vitro/functional, model-organism, and computational-database evidence. Ontology suggestions (MONDO:0013120, HGNC:1552, GO:0007530/0008584/0035102, UBERON:0000991/0000992, CL:0000216, NCIT clinical-intervention terms) are provided for knowledge-base ingestion.
